

Tutorial 5

Indian Institute of Technology Jodhpur
Mathematics-I (MAL1010)

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1. Check whether the following series are convergent or divergent.

a. $\sum_{n=1}^{\infty} \frac{1}{n(n+1)},$

b. $\sum_{n=1}^{\infty} \frac{n!}{n^n},$

c. $\sum_{n=1}^{\infty} \frac{1}{n^{1+\frac{1}{n}}},$

d. $\sum_{n=1}^{\infty} \frac{\log n}{n^\alpha}, \alpha > 1,$

e. $\sum_{n=1}^{\infty} \frac{n^2}{2^n},$

f. $\sum_{n=1}^{\infty} \frac{1}{2^{\log n}},$

g. $\sum_{n=1}^{\infty} \frac{b^n}{a^n + b^n}, 0 < a < b,$

h. $\sum_{n=1}^{\infty} \frac{n!}{1 \cdot 3 \cdot 5 \cdots (2n-1)},$

i. $\sum_{n=1}^{\infty} \left(\frac{1 \cdot 3 \cdot 5 \cdots (2n-3)}{2 \cdot 4 \cdot 6 \cdots (2n-2)} \right) \frac{1}{2n-1}.$

2. If $\{a_1, a_2, a_3, \dots\}$ be the collection of those natural number that ends with 1, prove that the series $\sum \frac{1}{a_n}$ is divergent.

3. Prove that if $x_n > 0, \forall n$, and $\sum_{n=1}^{\infty} x_n$ is convergent then $\sum_{n=1}^{\infty} \frac{\sqrt{x_n}}{n}$ is also convergent.

4. Prove that $\sum_{n=1}^{\infty} \frac{x_n}{1+n^2 x_n}$ is always convergent when $x_n > 0, \forall n$.

5. Prove that if $x_n > 0, \forall n$, and $\sum_{n=1}^{\infty} x_n$ is convergent, then $\sum_{n=1}^{\infty} e^{x_n}$ is divergent.

6. Prove that the series $\frac{a}{b} + \frac{a(a+c)}{b(b+c)} + \frac{a(a+c)(a+2c)}{b(b+c)(b+2c)} + \dots, a, b, c > 0$ is convergent if $b > a + c$ and divergent if $b \leq a + c$.

7. Show that the series $\frac{1}{(1+a)^p} - \frac{1}{(2+a)^p} + \frac{1}{(3+a)^p} - \dots, a > 0$ is

- (a) absolutely convergent if $p > 1$,
- (b) conditionally convergent if $0 < p \leq 1$.

8. If $\sum u_n$ be a convergent series, show that the following series are convergent.

a. $\sum \frac{u_n}{n},$

b. $\sum n^{\frac{1}{n}} u_n,$

c. $\sum \frac{u_n}{\log(n+1)}.$